

CS214: Computer Organization

Quiz 1 – Solutions

Maximum Marks: 15

MCQ Answer Key

Question	Q1	Q2	Q3	Q4	Q5
Answer	A	C	D	B	A

Q1.

[1 Mark]

For a T flip-flop,

$$Q^+ = Q \oplus T.$$

Starting from $Q_2Q_1Q_0 = 101$:

$$101 \rightarrow 110 \rightarrow 111 \rightarrow 000.$$

Hence, after three active clock edges,

$$Q_2Q_1Q_0 = 000.$$

Correct option: (A)

Q2.

[1 Mark]

Starting from 1011:

$$1011 \xrightarrow{\text{right shift}} 0101 \xrightarrow{\text{left shift}} 1010 \xrightarrow{\text{hold}} 1010 \xrightarrow{\text{right shift}} 0101.$$

Therefore, the final register contents are

$$0101.$$

Correct option: (C)

Q3.

[1 Mark]

Using

$$Q_3^+ = Q_2, \quad Q_2^+ = Q_1, \quad Q_1^+ = Q_0, \quad Q_0^+ = Q_3 \oplus Q_2,$$

the states are

$$\begin{aligned} 0001 &\rightarrow 0010 \rightarrow 0100 \rightarrow 1001 \rightarrow 0011 \\ &\rightarrow 0110 \rightarrow 1101 \rightarrow 1010 \rightarrow 0101 \\ &\rightarrow 1011 \rightarrow 0111 \rightarrow 1111 \rightarrow 1110 \\ &\rightarrow 1100 \rightarrow 1000 \rightarrow 0001. \end{aligned}$$

Thus, there are

$$15$$

distinct states, including the initial state, before the first return to 0001.

Correct option: (D)

Q4.

[1 Mark]

In an asynchronous (ripple) counter, the flip-flops do not change simultaneously. The output transition of one flip-flop triggers the next flip-flop, and every flip-flop has a nonzero propagation delay.

Therefore, during

$$0111 \rightarrow 1000,$$

the changes ripple through the stages one after another:

$$0111 \rightarrow 0110 \rightarrow 0100 \rightarrow 0000 \rightarrow 1000.$$

These are transient states caused by accumulated propagation delays; they are not additional stable count states.

Correct option: (B)

Q5.

[1 Mark]

Given

$$(G_3G_2G_1G_0) = 0100, \quad (P_3P_2P_1P_0) = 1011, \quad C_0 = 1,$$

and

$$C_{i+1} = G_i + P_iC_i,$$

we obtain

$$C_1 = 0 + (1)(1) = 1,$$

$$C_2 = 0 + (1)(1) = 1,$$

$$C_3 = 1 + (0)(1) = 1,$$

$$C_4 = 0 + (1)(1) = 1.$$

Hence,

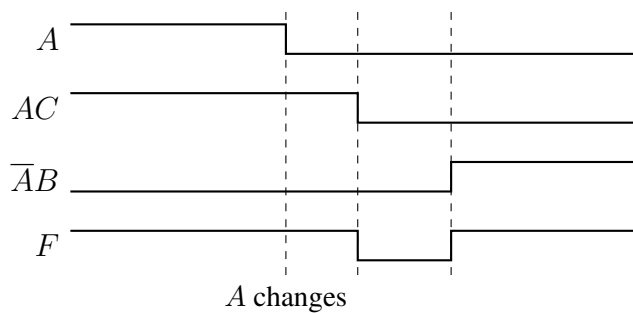
$$(C_1, C_2, C_3, C_4) = (1, 1, 1, 1).$$

Correct option: (A)

Q6.

[3 Marks]

Solution:



Thus, the output momentarily undergoes

$$1 \rightarrow 0 \rightarrow 1,$$

even though logically it should remain at 1.

(b)

Using the consensus theorem,

$$XY + \overline{X}Z + YZ = XY + \overline{X}Z,$$

the consensus term for

$$\overline{A}B + AC$$

is BC . Hence, an equivalent hazard-free realization is

$$F = \overline{A}B + AC + BC$$

The added term BC does not change the Boolean function.

During the given transition,

$$B = C = 1,$$

so

$$BC = 1$$

throughout. Therefore, even if both $\overline{A}B$ and AC are momentarily zero, the term BC keeps the output equal to 1, eliminating the static-1 hazard.

Q7.

[2 Marks]

For the register-to-register path, the setup constraint is

$$d_1 + t_p + t_{\text{comb}} \leq T + d_2 - t_{\text{setup}}.$$

Therefore,

$$T_{\min} = t_p + t_{\text{comb}} + t_{\text{setup}} + d_1 - d_2.$$

For Design I,

$$d_1 = 3 \text{ ns}, \quad d_2 = 5 \text{ ns},$$

so

$$T_{\min} = 2 + 6 + 1 + 3 - 5 = \boxed{7 \text{ ns}}.$$

For Design II,

$$d_1 = 5 \text{ ns}, \quad d_2 = 3 \text{ ns},$$

so

$$T_{\min} = 2 + 6 + 1 + 5 - 3 = \boxed{11 \text{ ns}}.$$

Thus, the later arrival of the clock at the receiving flip-flop in Design I provides positive clock skew and relaxes the setup-time constraint, whereas Design II makes the setup constraint more restrictive.

Q8.

[5 Marks] Q8.

[5 Marks]

Let the two inputs of the final cross-coupled NAND latch be denoted by $\overline{S}$ and $\overline{R}$.

Since the final SR latch is implemented using NAND gates, its inputs are active LOW:

$\overline{S}$	$\overline{R}$	Operation
1	1	Hold
0	1	Set ($Q = 1$)
1	0	Reset ($Q = 0$)

(a) Initially,

$$CLK = 0, \quad D = 1.$$

When $CLK = 0$, the clock-controlled NAND gates have a LOW clock input. Therefore, their outputs are HIGH. After the complete internal network has settled, the inputs of the final NAND SR latch are

$$\boxed{\overline{S} = 1, \quad \overline{R} = 1}.$$

Thus, the final SR latch is initially in the **hold** condition.

The internal feedback network, however, establishes the appropriate stable internal values corresponding to

$$D = 1.$$

Now let the clock make a positive transition,

$$CLK : 0 \rightarrow 1.$$

Since $D = 1$ was present before the positive clock edge, the internal NAND network changes the inputs of the final SR latch to

$$\boxed{\overline{S} = 0, \quad \overline{R} = 1}.$$

This is the active-low **set** condition. Therefore,

$$\boxed{Q = 1, \quad \overline{Q} = 0}.$$

Hence, the value

$$\boxed{D = 1}$$

present at the positive clock edge is stored.

Notice that while CLK remains HIGH, there is no requirement for $\overline{S}$ to return immediately to 1. For the present internal state of this circuit,

$$\boxed{\overline{S} = 0, \quad \overline{R} = 1}$$

is maintained, which continues to enforce

$$Q = 1.$$

(b) Now keep

$$CLK = 1$$

and change the input from

$$D : 1 \rightarrow 0.$$

Before this change, from part (a),

$$\overline{S} = 0, \quad \overline{R} = 1,$$

and therefore

$$Q = 1, \quad \overline{Q} = 0.$$

Although D now changes to 0, the internal feedback network has already established the state corresponding to the positive clock edge. The change in D does not create a reset condition while CLK remains HIGH.

Therefore, after the internal signals settle,

$$\boxed{\overline{S} = 0, \quad \overline{R} = 1}.$$

Thus, the final NAND SR latch is still in the set condition and

$$\boxed{Q = 1}$$

remains unchanged.

Hence,

$$\boxed{D : 1 \rightarrow 0 \quad \text{while} \quad CLK = 1}$$

does not change the output.

The important point is that the change in D after the positive edge cannot change the effective SR inputs during the same HIGH clock level. The value captured at the clock edge therefore remains stored.

(c) Part (b) shows the edge-triggered nature of the circuit.

After the positive clock edge, changing

$$D : 1 \rightarrow 0$$

while

$$CLK = 1$$

does not change the effective inputs of the final SR latch:

$$\boxed{(\overline{S}, \overline{R}) = (0, 1)}.$$

Consequently,

$$\boxed{Q = 1}$$

remains unchanged.

Thus, once the value has been captured at the positive clock edge, subsequent changes in D during the same HIGH clock level do not propagate to the output.

In a master-slave implementation, one latch is transparent during an entire clock level. Hence, a temporary unwanted input value appearing during the transparent interval may be captured. This is referred to as **0/1 catching**.

The behavior demonstrated in part (b) shows why the edge-triggered circuit avoids this problem: a change in D after the active edge, while the clock remains HIGH, does not alter the stored value.

The input must still satisfy the required setup- and hold-time conditions around the active clock edge.